

Atividade-Aula08

The screenshot shows the pgAdmin 4 interface with the following components:

- Browser:** Displays the database structure for 'PostgreSQL 14', including 'matht_for_comp_sci', 'postgres', and 'sistema_bancario'.
- Query Editor:** Contains SQL queries for creating tables and inserting data.

```
12
13 CREATE TABLE contas (
14     id_conta SERIAL PRIMARY KEY,
15     numero_conta VARCHAR(18) UNIQUE NOT NULL,
16     saldo DECIMAL(10,2) DEFAULT 0,
17     id_cliente INT REFERENCES clientes(id_cliente) ON DELETE CASCADE
18 );
19
20 INSERT INTO contas (numero_conta, saldo, id_cliente) VALUES
21 ('000123', 1500.00, 1),
22 ('000456', 2300.00, 2);
23
24 CREATE TABLE transacoes (
25     id_transacao SERIAL PRIMARY KEY,
26     id_conta INT REFERENCES contas(id_conta) ON DELETE CASCADE,
27     tipo VARCHAR(15) CHECK (tipo IN ('Depósito', 'Saque', 'Transferência')),
28     valor DECIMAL(10,2) NOT NULL CHECK (valor > 0),
29     data_transacao TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
30     destino_transferencia INT REFERENCES contas(id_conta)
31 );
32
33 INSERT INTO transacoes (id_conta, tipo, valor) VALUES
34 (1, 'Depósito', 500.00),
35 (2, 'Saque', 200.00),
36 (1, 'Transferência', 300.00);
37
38 SELECT * FROM clientes;
39
```
- Data Output:** Displays the results of the 'SELECT * FROM clientes;' query.

id_cliente [PK] integer	nome character varying (100)	cpf character varying (11)	endereco text	telefone character varying (15)
1	João Silva	12345678900	Rua A, 123	11999990000
2	Maria Oliveira	98765432100	Rua B, 456	11988887777

The screenshot shows the pgAdmin 4 interface with the following components:

- Browser:** Displays the database structure for 'PostgreSQL 14', including 'matht_for_comp_sci', 'postgres', and 'sistema_bancario'.
- Query Editor:** Contains SQL queries for creating tables, inserting data, and joining tables.

```
16     saldo DECIMAL(10,2) DEFAULT 0,
17     id_cliente INT REFERENCES clientes(id_cliente) ON DELETE CASCADE
18 );
19
20 INSERT INTO contas (numero_conta, saldo, id_cliente) VALUES
21 ('000123', 1500.00, 1),
22 ('000456', 2300.00, 2);
23
24 CREATE TABLE transacoes (
25     id_transacao SERIAL PRIMARY KEY,
26     id_conta INT REFERENCES contas(id_conta) ON DELETE CASCADE,
27     tipo VARCHAR(15) CHECK (tipo IN ('Depósito', 'Saque', 'Transferência')),
28     valor DECIMAL(10,2) NOT NULL CHECK (valor > 0),
29     data_transacao TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
30     destino_transferencia INT REFERENCES contas(id_conta)
31 );
32
33 INSERT INTO transacoes (id_conta, tipo, valor) VALUES
34 (1, 'Depósito', 500.00),
35 (2, 'Saque', 200.00),
36 (1, 'Transferência', 300.00);
37
38 SELECT * FROM clientes;
39
40 SELECT contas.numero_conta, contas.saldo, clientes.nome, clientes.cpf
41 FROM contas
42 JOIN clientes ON contas.id_cliente = clientes.id_cliente;
43
```
- Data Output:** Displays the results of the 'SELECT * FROM clientes;' query.

numero_conta character varying (10)	saldo numeric (10,2)	nome character varying (100)	cpf character varying (11)
1 000123	1500.00	João Silva	12345678900
2 000456	2300.00	Maria Oliveira	98765432100

pgAdmin 4

File Object Tools Help

Browser

- Servers (1)
 - PostgreSQL 14
 - Databases (3)
 - math_for_comp_sci
 - postgres
 - sistema_bancario**
 - Casts
 - Catalogs
 - Event Triggers
 - Extensions
 - Foreign Data Wrappers
 - Languages
 - Publications
 - Schemas
 - Subscriptions
 - Login/Group Roles
 - Tablespaces

Dashboard Properties SQL Statistics Dependencies Dependents sistema_bancario/postgres@PostgreSQL 14 *

Query Editor Query History

```
18 saldo DECIMAL(10,2) DEFAULT 0,
19 id_cliente INT REFERENCES clientes(id_cliente) ON DELETE CASCADE
20 );
21
22 INSERT INTO contas (numero_conta, saldo, id_cliente) VALUES
23 ('000123', 1500.00, 1),
24 ('000456', 2300.00, 2);
25
26 CREATE TABLE transacoes (
27 id_transacao SERIAL PRIMARY KEY,
28 id_conta INT REFERENCES contas(id_conta) ON DELETE CASCADE,
29 tipo VARCHAR(15) CHECK (tipo IN ('Depósito', 'Saque', 'Transferência')),
30 valor DECIMAL(10,2) NOT NULL CHECK (valor > 0),
31 data_transacao TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
32 destino_transferencia INT REFERENCES contas(id_conta)
33 );
34
35 INSERT INTO transacoes (id_conta, tipo, valor) VALUES
36 (1, 'Depósito', 500.00),
37 (2, 'Saque', 200.00),
38 (1, 'Transferência', 300.00);
39
40 SELECT * FROM clientes;
41
42 SELECT contas.numero_conta, contas.saldo, clientes.nome, clientes.cpf
43 FROM contas
44 JOIN clientes ON contas.id_cliente = clientes.id_cliente;
45
```

Data Output Explain Messages Notifications

id_cliente	nome	cpf	endereco	telefone
PK Integer	character varying (100)	character varying (11)	text	character varying (15)
1	João Silva	12345678900	Rua A, 123	11999990000
2	Maria Oliveira	98765432100	Rua B, 456	11988887777
3	Carlos Pereira	11223344556	Rua C, 789	11977776666

pgAdmin 4

File Object Tools Help

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Dashboard Properties SQL Statistics Dependencies Dependents sistema_bancario/postgres@PostgreSQL 14 *

Query Editor Query History

```
20 );
21
22 INSERT INTO contas (numero_conta, saldo, id_cliente) VALUES
23 ('000123', 1500.00, 1),
24 ('000456', 2300.00, 2),
25 ('000789', 0, 3);
26
27 CREATE TABLE transacoes (
28 id_transacao SERIAL PRIMARY KEY,
29 id_conta INT REFERENCES contas(id_conta) ON DELETE CASCADE,
30 tipo VARCHAR(15) CHECK (tipo IN ('Depósito', 'Saque', 'Transferência')),
31 valor DECIMAL(10,2) NOT NULL CHECK (valor > 0),
32 data_transacao TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
33 destino_transferencia INT REFERENCES contas(id_conta)
34 );
35
36 INSERT INTO transacoes (id_conta, tipo, valor) VALUES
37 (1, 'Depósito', 500.00),
38 (2, 'Saque', 200.00),
39 (1, 'Transferência', 300.00);
40
41 SELECT * FROM clientes;
42
43 SELECT contas.numero_conta, contas.saldo, clientes.nome, clientes.cpf
44 FROM contas
45 JOIN clientes ON contas.id_cliente = clientes.id_cliente;
46
47
```

Data Output Explain Messages Notifications

numero_conta	saldo	nome	cpf
character varying (10)	numeric (10,2)	character varying (100)	character varying (11)
1 000123	1500.00	João Silva	12345678900
2 000456	2300.00	Maria Oliveira	98765432100
3 000789	0.00	Carlos Pereira	11223344556

pgAdmin 4

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Dashboard Properties SQL Statistics Dependencies Dependents sistema_bancario/postgres@PostgreSQL 14*

Query Editor Query History

```
44 SELECT contas.numero_conta, contas.saldo, clientes.nome, clientes.cpf
45 FROM contas
46 JOIN clientes ON contas.id_cliente = clientes.id_cliente;
47
48 INSERT INTO transacoes (id_conta, tipo, valor)
49 VALUES (
50     (SELECT id_conta FROM contas WHERE numero_conta = '000123'),
51     'Saque',
52     100.00
53 );
54
55 INSERT INTO transacoes (id_conta, tipo, valor)
56 VALUES (
57     (SELECT id_conta FROM contas WHERE numero_conta = '000789'),
58     'Depósito',
59     100.00
60 );
61
62 UPDATE contas
63 SET saldo = saldo - 100.00
64 WHERE numero_conta = '000123';
65
66 UPDATE contas
67 SET saldo = saldo + 100.00
68 WHERE numero_conta = '000789';
69
70
```

Data Output Explain Messages Notifications

UPDATE 1

Query returned successfully in 42 msec.

Windows taskbar: Pesquisa, 22:40, 17/03/2026

pgAdmin 4

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Dashboard Properties SQL Statistics Dependencies Dependents sistema_bancario/postgres@PostgreSQL 14*

Query Editor Query History

```
46 JOIN clientes ON contas.id_cliente = clientes.id_cliente;
47
48 INSERT INTO transacoes (id_conta, tipo, valor)
49 VALUES (
50     (SELECT id_conta FROM contas WHERE numero_conta = '000123'),
51     'Saque',
52     100.00
53 );
54
55 INSERT INTO transacoes (id_conta, tipo, valor)
56 VALUES (
57     (SELECT id_conta FROM contas WHERE numero_conta = '000789'),
58     'Depósito',
59     100.00
60 );
61
62 UPDATE contas
63 SET saldo = saldo - 100.00
64 WHERE numero_conta = '000123';
65
66 UPDATE contas
67 SET saldo = saldo + 100.00
68 WHERE numero_conta = '000789';
69
70 SELECT contas.numero_conta, contas.saldo, clientes.nome
71 FROM contas
72 JOIN clientes ON contas.id_cliente = clientes.id_cliente
73 ORDER BY contas.numero_conta;
```

Data Output Explain Messages Notifications

numero_conta	saldo	nome
1 000123	1400.00	Jóão Silva
2 000456	2300.00	Maria Oliveira
3 000789	100.00	Carlos Pereira

Windows taskbar: Pesquisa, 22:42, 17/03/2026

```
CREATE TABLE clientes (  
id_cliente SERIAL PRIMARY KEY,  
nome VARCHAR(100) NOT NULL,  
cpf VARCHAR(11) UNIQUE NOT NULL,  
endereco TEXT,  
telefone VARCHAR(15)  
);
```

```
INSERT INTO clientes (nome, cpf, endereco, telefone) VALUES  
(  
'João Silva', '12345678900', 'Rua A, 123', '11999990000'),  
(  
'Maria Oliveira', '98765432100', 'Rua B, 456', '11988887777'),  
(  
'Carlos Pereira', '11223344556', 'Rua C, 789', '11977776666');
```

```
CREATE TABLE contas (  
id_conta SERIAL PRIMARY KEY,  
numero_conta VARCHAR(10) UNIQUE NOT NULL,  
saldo DECIMAL(10,2) DEFAULT 0,  
id_cliente INT REFERENCES clientes(id_cliente) ON DELETE CASCADE  
);
```

```
INSERT INTO contas (numero_conta, saldo, id_cliente) VALUES  
(  
'000123', 1500.00, 1),  
(  
'000456', 2300.00, 2),  
(  
'000789', 0, 3);
```

```
CREATE TABLE transacoes (  
id_transacao SERIAL PRIMARY KEY,  
id_conta INT REFERENCES contas(id_conta) ON DELETE CASCADE,  
tipo VARCHAR(15) CHECK (tipo IN ('Depósito', 'Saque', 'Transferência')),
```

```
valor DECIMAL(10,2) NOT NULL CHECK (valor > 0),  
data_transacao TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
destino_transferencia INT REFERENCES contas(id_conta)  
);
```

```
INSERT INTO transacoes (id_conta, tipo, valor) VALUES  
(1, 'Depósito', 500.00),  
(2, 'Saque', 200.00),  
(1, 'Transferência', 300.00);
```

```
SELECT * FROM clientes;
```

```
SELECT contas.numero_conta, contas.saldo, clientes.nome, clientes.cpf  
FROM contas  
JOIN clientes ON contas.id_cliente = clientes.id_cliente;
```

```
INSERT INTO transacoes (id_conta, tipo, valor)  
VALUES (  
(SELECT id_conta FROM contas WHERE numero_conta = '000123'),  
'Saque',  
100.00  
);
```

```
INSERT INTO transacoes (id_conta, tipo, valor)  
VALUES (  
(SELECT id_conta FROM contas WHERE numero_conta = '000789'),  
'Depósito',  
100.00  
);
```

```
UPDATE contas
```

```
SET saldo = saldo - 100.00
```

```
WHERE numero_conta = '000123';
```

```
UPDATE contas
```

```
SET saldo = saldo + 100.00
```

```
WHERE numero_conta = '000789';
```

```
SELECT contas.numero_conta, contas.saldo, clientes.nome
```

```
FROM contas
```

```
JOIN clientes ON contas.id_cliente = clientes.id_cliente
```

```
ORDER BY contas.numero_conta;
```